

Course title: Handling Structured Data (Tables → Models → Pipelines)

Challenge Worksheet

Build an End-to-End Structured Data Workflow

What this challenge is about

Structured data powers most business analytics and machine learning systems. In this challenge, you will design and document a complete workflow that takes a raw tabular dataset, cleans it, transforms it, trains a model, and prepares the process for automation using pipelines.

You will work on the topics that we presented in our session:

- Working with tables
- Training a simple ML model on structured data
- Automating the workflow using pipelines

Important note: For a simplified version of the exercise, please see below. The exercise is designed to cover all concepts covered during the session, but for learners that might be more new to the topic, the simplified version is presented as well.

Your Objective

Design the workflow using python on Visual Studio Code.

Why this matters

Machine learning with structured data only becomes reliable and scalable when the workflow is consistent and automated. Pipelines reduce manual errors, enforce repeatability, and make it possible to retrain models on new data without rebuilding everything from scratch.

By completing this challenge, you will understand how to turn a manual ML workflow into a production-ready process

The Challenge

Follow the steps below to design your structured data pipeline.

Use the iris.csv dataset which can be downloaded [here](#):

Step 1: Install the environment

Step 2: Identify your dataset and define your goal. Think about data types.

Step 3: Select your ML model. Use AI assistance if needed.

Step 4: Build the pipeline that automates preprocessing and training. You can use the notebook presented in the session as a template. Use AI tools for guidance

Step 5: Define how to evaluate the model and how the pipeline could be scheduled

Submission Checklist

Before uploading, make sure your file includes:

- Jupyter notebook file where the pipeline is created
- Your solution should be analogous to the notebook “pipelines with structured data.ipynb” which you will find in the resources provided at the end of this document. You should use the iris dataset
- Data should be loaded, transformed and the data model should be built before packaging into a pipeline
- You can use AI for assistance if you need support

! *Once submitted, solutions cannot be edited or resubmitted. Please take a moment to review before uploading.*

Practical Walkthrough

Not sure where to start? Follow this walkthrough step by step.

Step-by-Step Guide

1. Load and explore your dataset

- Open VS Code
- Create a new Python notebook
- Load your CSV using pandas
- Inspect the first rows, data types, and missing values

2. Clean and transform the data

- Fill missing values using median or most frequent
- Create new features

3. Split into train/test sets

- Use `train_test_split` to ensure fair evaluation

4. Build your preprocessing pipeline

- Use `ColumnTransformer` to separate numeric and categorical features
- Add imputers, scalers, and pipelines

5. Add your ML model

- Start with a simple ML model
- Wrap everything in a Pipeline

6. Train and evaluate

- Fit the pipeline on training data

7. Think about automation

- How would you retrain this pipeline weekly?
- How would you ingest new data automatically?
- What outputs would you publish (predictions, metrics, logs)?

Simplified Version of the exercise

For a simpler approach that will accommodate beginners, you can simplify this exercise by using the same exercise that was presented in the class (Available here: [trainerclara/structured data](https://trainerclara.com/structured-data) and simply submit the exploration of alternative data transformations) For this version of the challenge, the goal would be to explore other data transformations before the model and pipeline objects are created. You can use the provided notebook as an example, without the need to modify the model and pipeline.

For further guidance, you can check the examples on:

[trainerclara/structured-data](https://trainerclara.com/structured-data)

Resources to set up the environment:

<https://code.visualstudio.com/>

<https://www.python.org/downloads/>

Tutorial to use python withing VS Code: <https://code.visualstudio.com/docs/languages/python>